

Part 1: Create the Publisher Script

[Sensor.py](#)

```
import paho.mqtt.client as mqtt
import json
import time
import random
from datetime import datetime, timezone

# MQTT setup
BROKER = "localhost"
PORT = 1883
TOPIC = "hydrocient/grandmarina/sensors/main-building/readings"

# Create client
client = mqtt.Client(mqtt.CallbackAPIVersion.VERSION2)

print("Starting device: GM-HYDROLOGIC-01")
print("Location: main-building")
print(f"Publishing to: {TOPIC}")
print("Interval: 2 seconds")
print("-----")

client.connect(BROKER, PORT)

counter = 1

while True:
    # Generate realistic values
    pressure_up = round(random.uniform(80, 85), 1)
    pressure_down = round(random.uniform(75, 80), 1)
    flow = round(random.uniform(38, 45), 1)

    # Create payload
    data = {
        "device_id": "GM-HYDROLOGIC-01",
        "location": "main-building",
        "timestamp": datetime.now(timezone.utc).isoformat(),
```

```

"counter": counter,
"pressure_upstream": pressure_up,
"pressure_downstream": pressure_down,
"flow_rate": flow
}

# Publish
client.publish(TOPIC, json.dumps(data))

# Print output
print(f"[{counter}] Pressure: {pressure_up}/{pressure_down} PSI, Flow: {flow}
gal/min")

counter += 1
time.sleep(2)

```

```

...documents/IoT-Externship/hydroficient-project -- Python dashboard_subscriber.py.txt
~/Documents/IoT-Externship/hydroficient-project -- Python sensor.py
~/Documents/IoT-Externship/hydroficient-project -- mosquito -v

Last login: Sun Mar 29 14:04:43 on ttys000
kaylacolletti@Mac hydroficient-project % python3 sensor_publisher.py
File ~/Users/kaylacolletti/Documents/IoT-Externship/hydroficient-project/sensor_publisher.py, line 1
(\rtf1\ansi\ansicpg1252\cocoartf2867
^
SyntaxError: unexpected character after line continuation character
kaylacolletti@Mac hydroficient-project % python3 sensor.py
Starting device: 0M-HYDROLOGIC-01
Location: main-building
Publishing to: hydroficient/grandmarina/sensors/main-building/readings
Interval: 2 seconds
-----
[1] Pressure: 80.2/77.2 PSI, Flow: 40.9 gal/min
[2] Pressure: 84.2/77.1 PSI, Flow: 41.0 gal/min
[3] Pressure: 81.4/79.6 PSI, Flow: 38.6 gal/min
[4] Pressure: 81.9/79.0 PSI, Flow: 39.9 gal/min
[5] Pressure: 82.3/75.3 PSI, Flow: 42.0 gal/min
[6] Pressure: 82.1/79.2 PSI, Flow: 42.4 gal/min
[7] Pressure: 82.9/78.1 PSI, Flow: 42.7 gal/min
[8] Pressure: 80.3/78.5 PSI, Flow: 42.9 gal/min
[9] Pressure: 81.9/76.5 PSI, Flow: 40.8 gal/min
[10] Pressure: 83.9/75.5 PSI, Flow: 42.7 gal/min
[11] Pressure: 84.6/76.2 PSI, Flow: 42.2 gal/min
[12] Pressure: 82.1/78.3 PSI, Flow: 39.6 gal/min
[13] Pressure: 81.7/76.7 PSI, Flow: 43.6 gal/min
[14] Pressure: 81.2/78.8 PSI, Flow: 44.5 gal/min
[15] Pressure: 80.5/77.6 PSI, Flow: 40.0 gal/min
[16] Pressure: 82.4/77.4 PSI, Flow: 38.1 gal/min
[17] Pressure: 83.5/77.5 PSI, Flow: 39.9 gal/min
[18] Pressure: 80.2/76.9 PSI, Flow: 41.0 gal/min
[19] Pressure: 83.2/75.8 PSI, Flow: 44.8 gal/min
[20] Pressure: 82.8/75.2 PSI, Flow: 42.5 gal/min
[21] Pressure: 83.5/76.6 PSI, Flow: 42.1 gal/min
[22] Pressure: 84.6/76.5 PSI, Flow: 44.2 gal/min
[23] Pressure: 84.9/79.2 PSI, Flow: 42.7 gal/min
[24] Pressure: 84.4/77.0 PSI, Flow: 42.4 gal/min
[25] Pressure: 82.3/76.3 PSI, Flow: 41.8 gal/min
[26] Pressure: 80.1/77.5 PSI, Flow: 43.9 gal/min
[27] Pressure: 84.3/76.4 PSI, Flow: 39.4 gal/min
[28] Pressure: 84.3/76.8 PSI, Flow: 40.2 gal/min
[29] Pressure: 84.2/75.7 PSI, Flow: 41.4 gal/min
[30] Pressure: 83.0/78.8 PSI, Flow: 42.8 gal/min
[31] Pressure: 82.7/77.3 PSI, Flow: 43.8 gal/min
[32] Pressure: 81.3/75.5 PSI, Flow: 43.4 gal/min
[33] Pressure: 84.3/75.9 PSI, Flow: 42.5 gal/min
[34] Pressure: 81.8/77.4 PSI, Flow: 40.7 gal/min
[35] Pressure: 80.9/75.6 PSI, Flow: 39.2 gal/min
[36] Pressure: 82.9/76.8 PSI, Flow: 43.6 gal/min
[37] Pressure: 80.5/79.0 PSI, Flow: 40.0 gal/min
[38] Pressure: 81.7/77.4 PSI, Flow: 38.6 gal/min
[39] Pressure: 83.2/76.0 PSI, Flow: 44.2 gal/min
[40] Pressure: 85.0/78.1 PSI, Flow: 42.0 gal/min
[41] Pressure: 83.4/75.3 PSI, Flow: 44.0 gal/min
[42] Pressure: 83.6/78.9 PSI, Flow: 38.0 gal/min
[43] Pressure: 80.6/77.0 PSI, Flow: 43.7 gal/min
[44] Pressure: 83.0/75.5 PSI, Flow: 43.5 gal/min
[45] Pressure: 80.5/77.2 PSI, Flow: 43.3 gal/min
[46] Pressure: 84.0/79.3 PSI, Flow: 44.0 gal/min

```

Part 2: Create the Subscriber Script

Dashboard_subscriber.py

```
import paho.mqtt.client as mqtt
import json
from datetime import datetime

# MQTT setup
BROKER = "localhost"
PORT = 1883
TOPIC = "hydroficient/grandmarina/#" # subscribe to all Grand Marina topics

def on_connect(client, userdata, flags, reason_code, properties=None):
    print("\n" + "="*60)
    print(" GRAND MARINA WATER MONITORING DASHBOARD")
    print(" Connected at:", datetime.now().strftime("%Y-%m-%d %H:%M:%S"))
    print("="*60)
    client.subscribe(TOPIC)

def on_message(client, userdata, msg):
    try:
        data = json.loads(msg.payload.decode())
        display_reading(data)
    except json.JSONDecodeError:
        # Non-JSON message (maybe a command or alert)
        print(f"\n[RAW MESSAGE] {msg.topic}")
        print(f"    {msg.payload.decode()}")

def display_reading(data):
    """Format and display a sensor reading."""
    print(f"\n{'-'*40}")
    print(f" Location: {data.get('location', 'Unknown')}")
    print(f" Device ID: {data.get('device_id', 'Unknown')}")
    print(f" Time:    {data.get('timestamp', 'N/A')}")
    print(f" Count:   #{data.get('counter', 0)}")
    print(f"{'-'*40}")

# Pressure readings
up = data.get('pressure_upstream', 0)
```

```
down = data.get('pressure_downstream', 0)
print(f" Pressure (upstream): {up:6.1f} PSI")
print(f" Pressure (downstream): {down:6.1f} PSI")

# Pressure differential
diff = up - down
print(f" Pressure differential: {diff:6.1f} PSI")

# Flow rate
flow = data.get('flow_rate', 0)
print(f" Flow rate: {flow:6.1f} gal/min")

# Create MQTT client
client = mqtt.Client(mqtt.CallbackAPIVersion.VERSION2)
client.on_connect = on_connect
client.on_message = on_message

print("Connecting to broker...")
client.connect(BROKER, PORT)
client.loop_forever()
```

=====

GRAND MARINA WATER MONITORING DASHBOARD

Connected at: 2026-03-29 14:06:57

=====

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:06:58.340920+00:00

Count: #17

Pressure (upstream): 83.5 PSI

Pressure (downstream): 77.5 PSI

Pressure differential: 6.0 PSI

Flow rate: 39.9 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:00.342268+00:00

Count: #18

Pressure (upstream): 80.2 PSI

Pressure (downstream): 76.9 PSI

Pressure differential: 3.3 PSI

Flow rate: 41.0 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:02.347866+00:00

Count: #19

Pressure (upstream): 83.2 PSI

Pressure (downstream): 75.8 PSI

Pressure differential: 7.4 PSI

Flow rate: 44.8 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:04.353665+00:00

Count: #20

Pressure (upstream): 82.8 PSI

Pressure (downstream): 75.2 PSI

Pressure differential: 7.6 PSI

Flow rate: 42.5 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:06.358342+00:00

Count: #21

Pressure (upstream): 83.5 PSI

Pressure (downstream): 76.6 PSI

Pressure differential: 6.9 PSI

Flow rate: 42.1 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Count: #22

Pressure (upstream): 84.6 PSI

Pressure (downstream): 76.5 PSI

Pressure differential: 8.1 PSI

Flow rate: 44.2 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:10.368521+00:00

Count: #23

Pressure (upstream): 84.9 PSI

Pressure (downstream): 79.2 PSI

Pressure differential: 5.7 PSI

Flow rate: 42.7 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:12.371150+00:00

Count: #24

Pressure (upstream): 84.4 PSI

Pressure (downstream): 77.0 PSI

Pressure differential: 7.4 PSI

Flow rate: 42.4 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:14.376723+00:00

Count: #25

Pressure (upstream): 82.3 PSI

Pressure (downstream): 76.3 PSI

Pressure differential: 6.0 PSI

Flow rate: 41.8 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:16.381521+00:00

Count: #26

Pressure (upstream): 80.1 PSI

Pressure (downstream): 77.5 PSI

Pressure differential: 2.6 PSI

Flow rate: 43.9 gal/min

Location: main-building

Device ID: GM-HYDROLOGIC-01

Time: 2026-03-29T18:07:18.385438+00:00

Count: #27

Pressure (upstream): 84.3 PSI

Pressure (downstream): 76.4 PSI

Pressure differential: 7.9 PSI

Flow rate: 39.4 gal/min

Location: main-building

Part 3:

Mosquitto Terminal-

```
...documents/loT-Externship/hydroficient-project -- Python dashboard_subscriber.py.txt
~/Documents/loT-Externship/hydroficient-project -- Python sensor.py
~/Documents/loT-Externship/hydroficient-project -- mosquitto -v

Last login: Sun Mar 29 14:05:50 on ttys001
kayiacolletti@Mac hydroficient-project % mosquitto -v
1774807567: Info: running mosquitto as user: kayiacolletti.
1774807567: mosquitto version 2.1.2 starting
1774807567: Using default config.
1774807567: Bridge support available.
1774807567: Persistence support available.
1774807567: TLS support available.
1774807567: TLS-PSK support available.
1774807567: Websockets support available.
1774807567: Starting in local only mode. Connections will only be possible from clients running on this machine.
1774807567: Create a configuration file which defines a listener to allow remote access.
1774807567: For more details see https://mosquitto.org/documentation/authentication-methods/
1774807567: Opening ipv4 listen socket on port 1883.
1774807567: Opening ipv6 listen socket on port 1883.
1774807567: mosquitto version 2.1.2 running
1774807586: New connection from ::1:60448 on port 1883.
1774807586: New client connected from ::1:60448 as auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (p4, c1, k60).
1774807586: No will message specified.
1774807586: Sending CONNACK to auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (0, 0)
1774807586: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807588: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807590: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807592: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807594: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807596: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807598: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807600: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807602: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (200 bytes))
1774807604: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807606: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807608: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807610: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807612: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807614: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807616: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807617: New connection from ::1:60449 on port 1883.
1774807617: New client connected from ::1:60449 as auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (p4, c1, k60).
1774807617: No will message specified.
1774807617: Sending CONNACK to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (0, 0)
1774807617: Received SUBSCRIBE from auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF
1774807617: hydroficient/grandmarina/# (QoS 0)
1774807617: auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF 0 hydroficient/grandmarina/#
1774807617: Sending SUBACK to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF
1774807618: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807618: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807620: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807620: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807622: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807622: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807624: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807624: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807626: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807626: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807628: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807628: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807630: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807630: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807632: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807632: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807634: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807634: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807636: Received PUBLISH from auto-C5C03E33-ECE6-C48A-8888-2F2F00FAEC68 (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
1774807636: Sending PUBLISH to auto-14932CB2-9FDC-F05C-2411-4AA5F5EA82BF (d0, q0, r0, m0, 'hydroficient/grandmarina/sensors/main-building/readings', ... (201 bytes))
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